

Partial Derivatives and the Gradient

ANSWER KEY



First and second partial derivatives

- 1 For $f(x, y) = x^3y^2 - 4xy + y$, find:
- a** $f_x = 3x^2y^2 - 4y$ **b** $f_y = 2x^3y - 4x + 1$
c $f_{xx} = 6xy^2$ **d** $f_{xy} = 6x^2y - 4$
- 2 For $f(x, y) = e^{xy}\sin(x + y)$, find f_x using the product rule.
 $f_x = ye^{xy}\sin(x + y) + e^{xy}\cos(x + y)$.
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The gradient and directional derivatives

- 3 Find the gradient ∇f of $f(x, y) = x^2y + 3y^3$ at the point $(1, 2)$.
 $\nabla f = \langle 2xy, x^2 + 9y^2 \rangle$. At $(1, 2)$: $\nabla f(1, 2) = \langle 4, 37 \rangle$.
- 4 Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.
 $\nabla f = \langle 2x, 2y \rangle$, so $\nabla f(1, 1) = \langle 2, 2 \rangle$. Unit vector: $\hat{\mathbf{v}} = \langle \frac{3}{5}, \frac{4}{5} \rangle$. $D_{\mathbf{v}}f = \langle 2, 2 \rangle \cdot \langle \frac{3}{5}, \frac{4}{5} \rangle = \frac{6}{5} + \frac{8}{5} = \frac{14}{5}$.
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Tangent planes

- 5 Find the equation of the tangent plane to $z = f(x, y) = x^2 + y^2$ at the point $(1, 2, 5)$, using
 $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.
 $f_x = 2x$, $f_y = 2y$: at $(1, 2)$, $f_x = 2$, $f_y = 4$. Tangent plane: $z - 5 = 2(x - 1) + 4(y - 2)$, i.e.
 $z = 2x + 4y - 5$.
- 6 Find the equation of the tangent plane to $z = xy$ at the point $(2, 3, 6)$.
 $f_x = y$, $f_y = x$: at $(2, 3)$, $f_x = 3$, $f_y = 2$. Tangent plane: $z - 6 = 3(x - 2) + 2(y - 3)$, i.e. $z = 3x + 2y - 6$.
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Critical points of multivariable functions

- 7 Find the critical point(s) of $f(x, y) = x^2 + y^2 - 4x + 2y + 5$ by setting $f_x = 0$ and $f_y = 0$.
 $f_x = 2x - 4 = 0$ gives $x = 2$; $f_y = 2y + 2 = 0$ gives $y = -1$. Critical point: $(2, -1)$.
- 8 Verify that the critical point found above is a minimum, using the second derivative test with
 $D = f_{xx}f_{yy} - (f_{xy})^2$.
 $f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 0$: $D = 2(2) - 0 = 4 > 0$ and $f_{xx} = 2 > 0$, so $(2, -1)$ is a local minimum.
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The chain rule for multivariable functions

- 9 Let $z = x^2y$, where $x = t^2$ and $y = t^3$. Use the multivariable chain rule $\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$ to find $\frac{dz}{dt}$ at $t = 1$.
 $\frac{\partial z}{\partial x} = 2xy$, $\frac{\partial z}{\partial y} = x^2$; $\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2$. At $t = 1$: $x = 1$, $y = 1$. $\frac{dz}{dt} = 2(1)(1)(2) + 1(3) = 4 + 3 = 7$.
- 10 Let $w = x^2 + y^2 + z^2$, where $x = \cos t$, $y = \sin t$, $z = t$. Find $\frac{dw}{dt}$ using the chain rule.
 $\frac{\partial w}{\partial x} = 2x$, $\frac{\partial w}{\partial y} = 2y$, $\frac{\partial w}{\partial z} = 2z$; $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$, $\frac{dz}{dt} = 1$.
 $\frac{dw}{dt} = 2x(-\sin t) + 2y\cos t + 2z = -2\cos t\sin t + 2\sin t\cos t + 2t = 2t$ (the first two terms cancel).
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Maximum rate of increase

- 11 Find the direction in which $f(x, y) = x^2 + xy$ increases most rapidly at the point $(1, 2)$, and state the maximum rate of increase.
 $\nabla f = \langle 2x + y, x \rangle$; at $(1, 2)$: $\nabla f = \langle 4, 1 \rangle$. Direction of steepest increase: $\langle 4, 1 \rangle$ (or its unit vector). Maximum rate of increase: $|\nabla f| = \sqrt{16 + 1} = \sqrt{17}$.
- 12 Find the direction of steepest descent for $f(x, y) = x^2 + y^2$ at the point $(3, 4)$.
 $\nabla f = \langle 2x, 2y \rangle$; at $(3, 4)$: $\langle 6, 8 \rangle$. Steepest descent is the opposite direction: $\langle -6, -8 \rangle$ (pointing toward the origin, the minimum).
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Implicit partial differentiation

- 13 For the surface $x^2 + y^2 + z^2 = 9$, find $\frac{\partial z}{\partial x}$ by implicit differentiation (treating y as constant).
 $2x + 2z\frac{\partial z}{\partial x} = 0$, so $\frac{\partial z}{\partial x} = -\frac{x}{z}$.